Method:

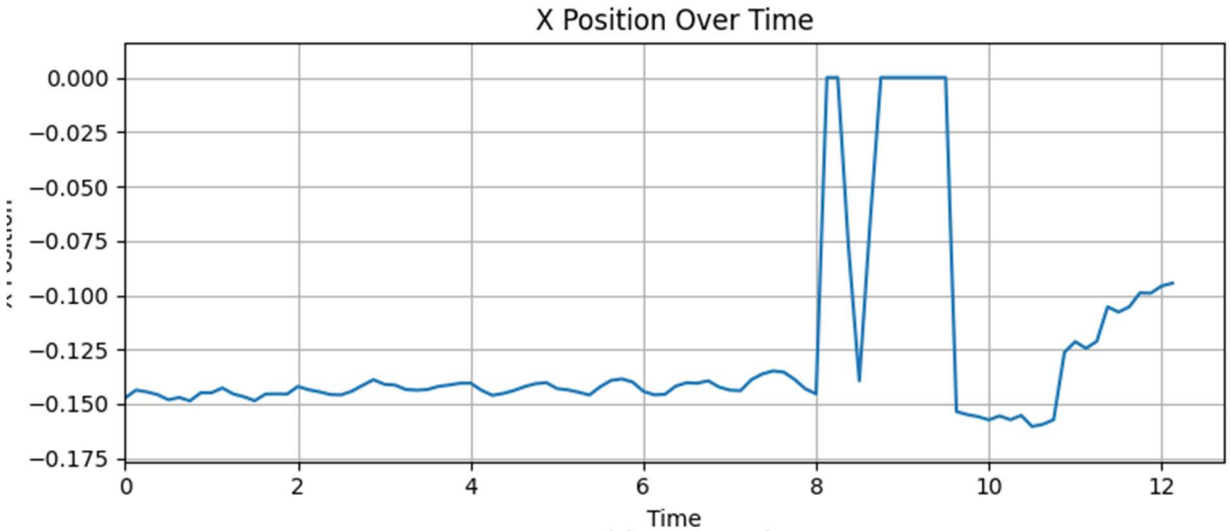
The current iteration of the prediction algorithm is simple. It allows room for additional modification and tuning to achieve more accurate results. Presently, the system takes a CSV file containing X, Y, Z position data of a selected point as obtained by the 3D camera on board the ROV. These values are fed one at a time (one per axis) to a prediction function which fills a circular buffer array of length N with the values. This is done to best suit integration for run-time prediction. When this buffer is not full, no prediction is made, once the buffers fill a fast Fourier transform is applied to the buffer recursively to compute the frequency spectrum of the latest data. Then, small signals, (below a percentage threshold of the dominant frequency's magnitude), are zeroed to remove noise from the system. Then an inverse fast Fourier transform is computed for n timesteps to predict the position of the point. Supporting functions display the frequency spectrum plots and calculate the L2-Norm of the error between the prediction made and the actual recorded position at that time. This allows for analysis of the algorithm's performance as adjustments are made.

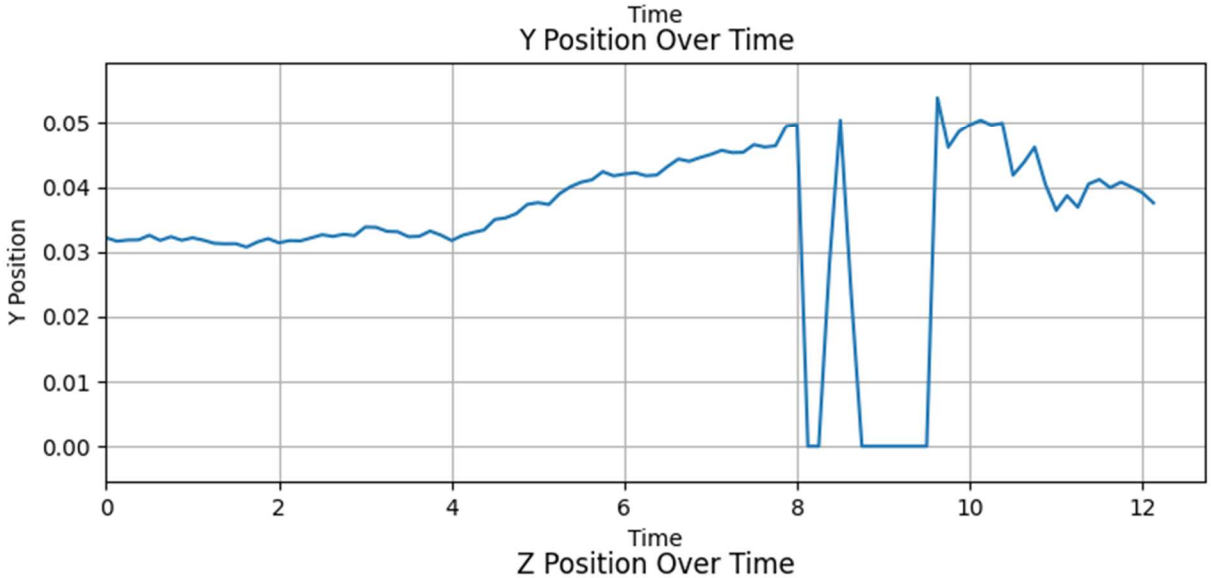
Data Set 1:

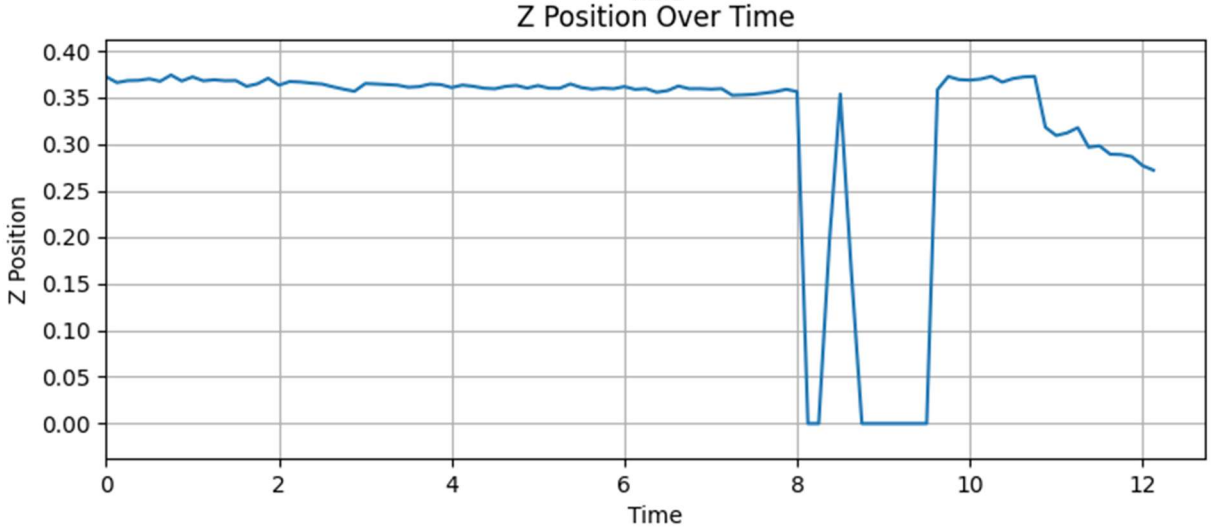
Num points: 98

Buffer Length: 50

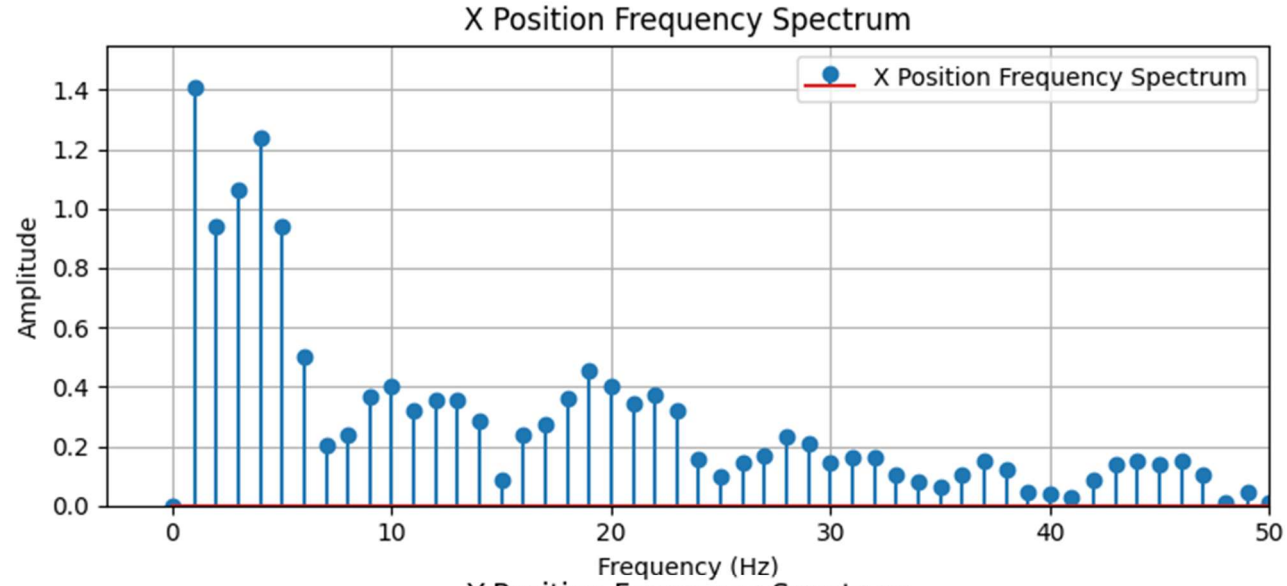
Input Per Axis:

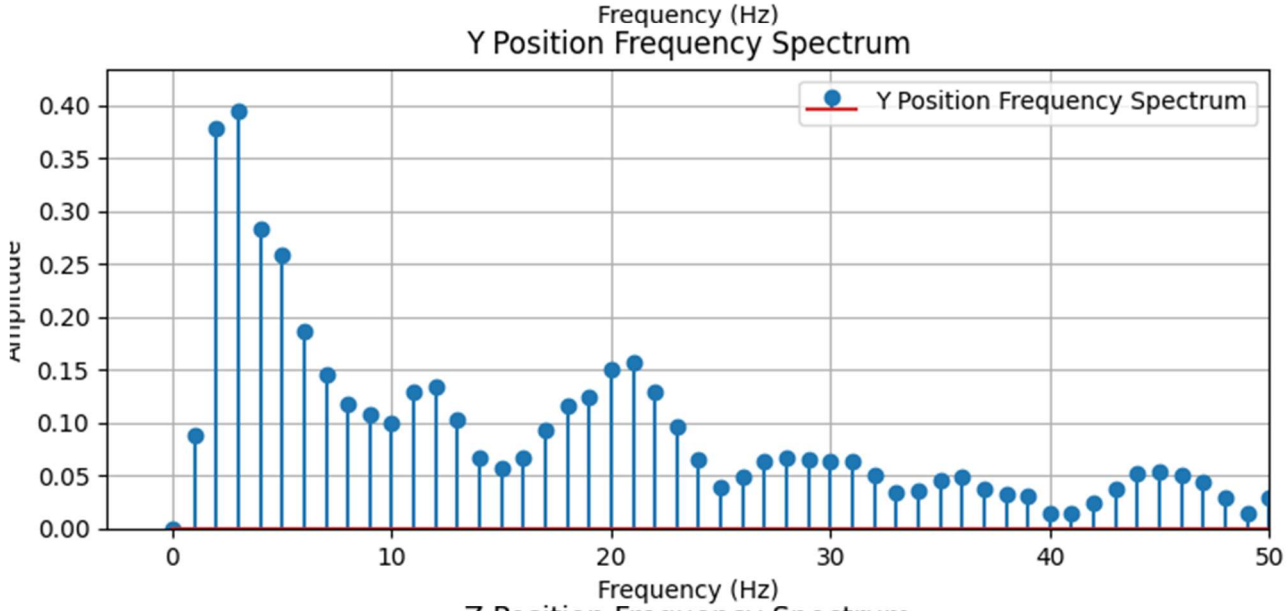


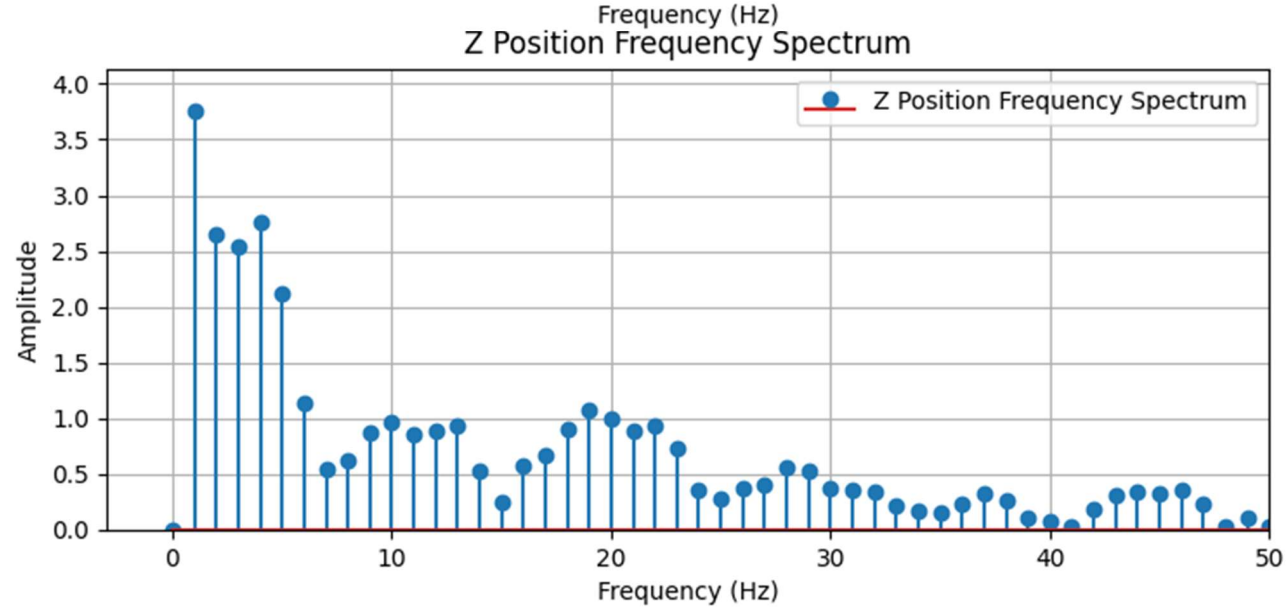




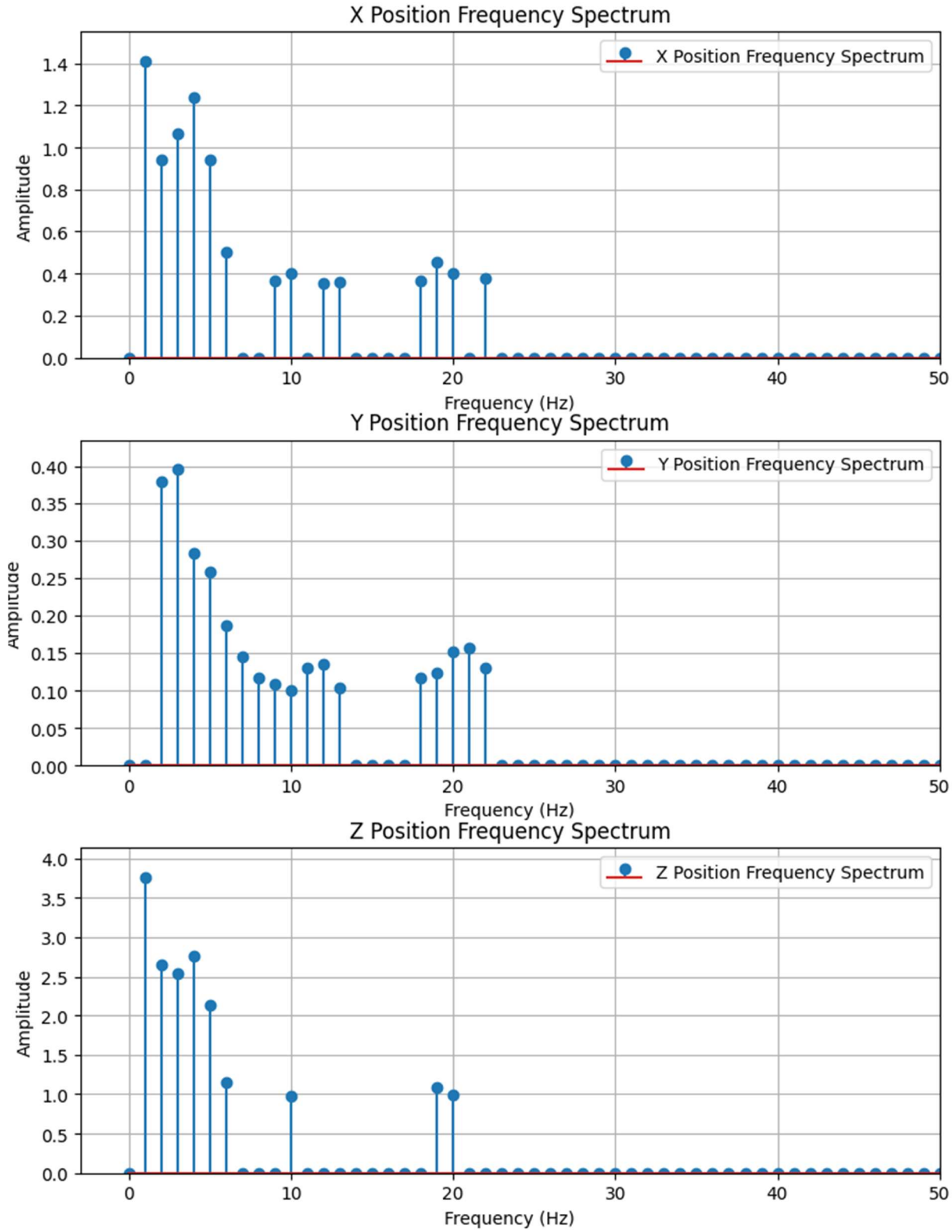
Input Frequency Spectrum:







Denoised Frequency Spectrum:



Observations:

This data is a relatively small set with only 98 points (assumed data interval of 0.2s or 5 points per second). Although it shows that there is a clear dominance from the lower frequencies, which is to be expected as both the ROV, and the tracked point moved slowly in the ocean. The frequency spectrum trails off until it fully diminishes at 50Hz. However, it can be seen that there are several small peaks approximately in gaps of approximately 9Hz as the frequency increases. After 'denoising' with a threshold of 25% most of these frequencies are diminished fully. Leaving only the strong frequencies around the 5Hz range. It can also be seen that X and Z axes have dominance at slightly lower frequencies, around 2-3 while Y is slightly higher at 4-6. This is not necessarily indicative of anything but potentially worth noting.

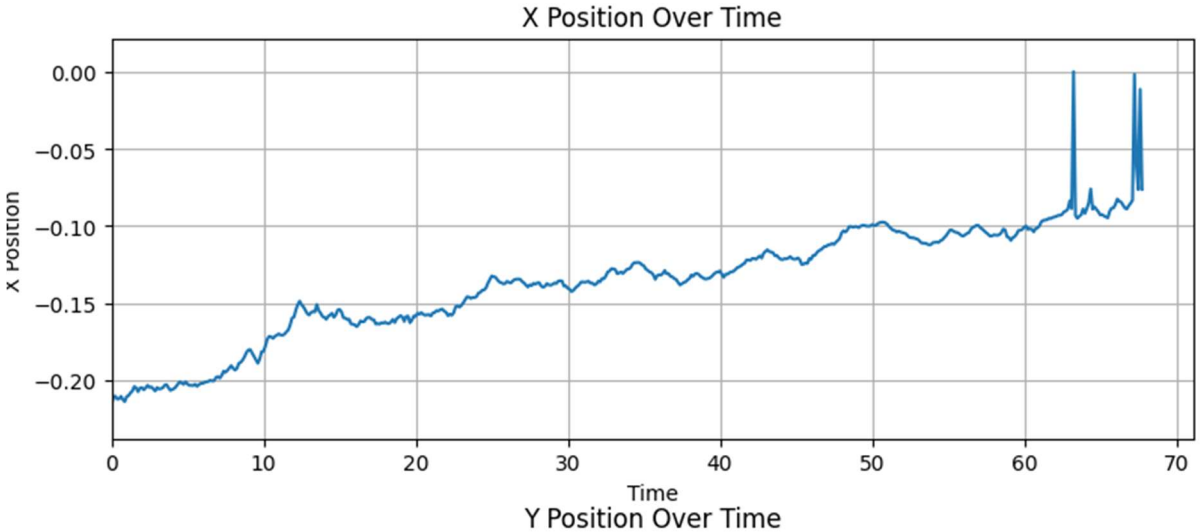
Average L2-Norm 1s Future of prediction: 0.33

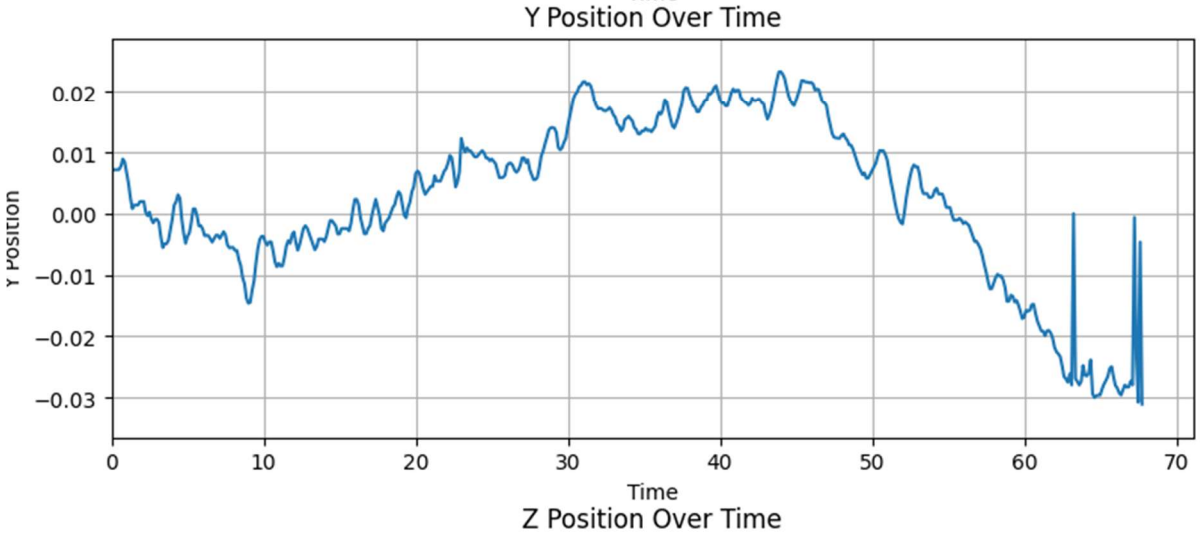
Data Set 2:

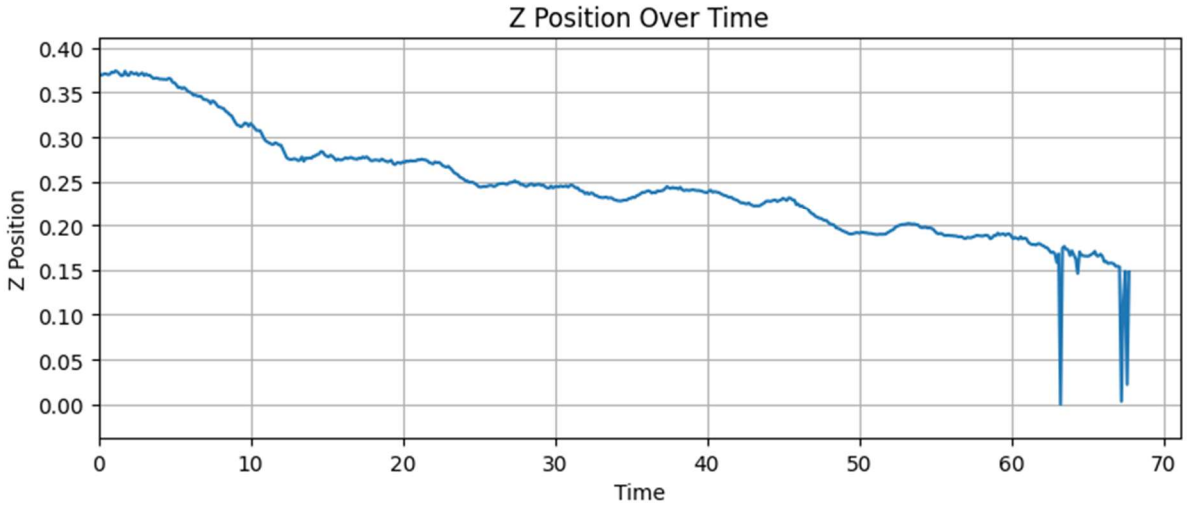
Num Points: 543

Buffer Length: 50

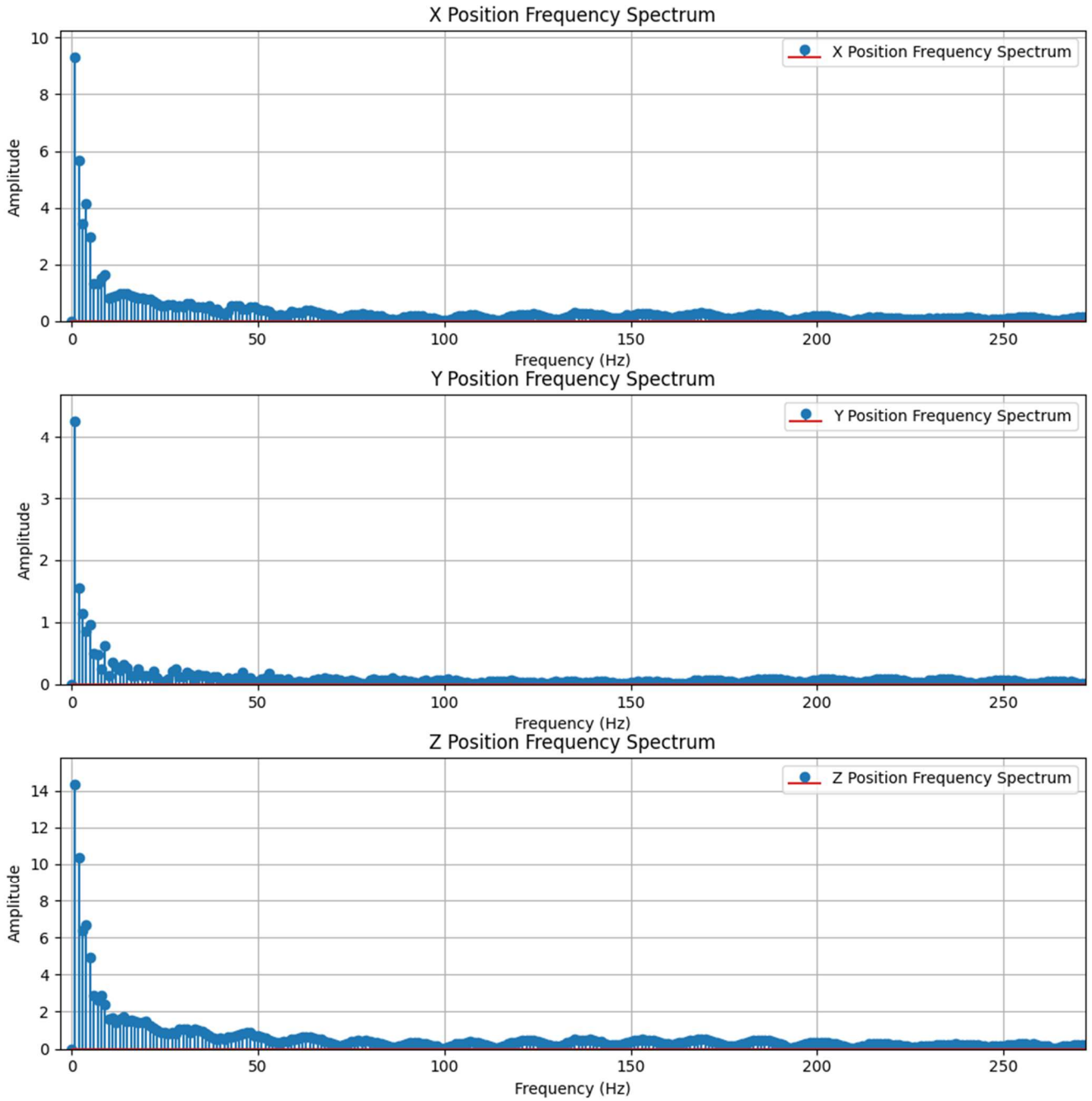
Input Per Axis:



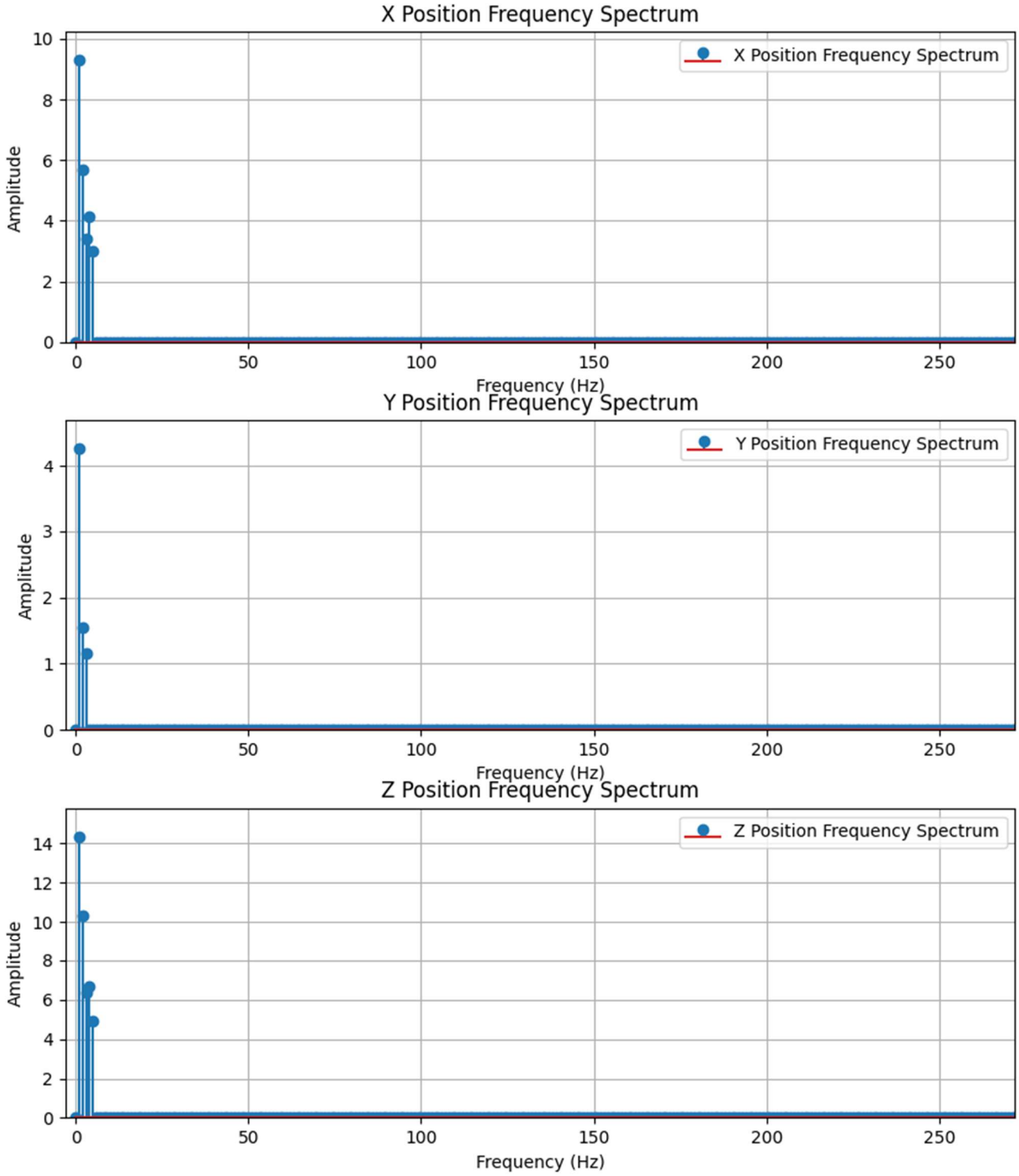




Input Frequency Spectrum:



Denoised Frequency Spectrum:



Observations:

Dataset 2 has notably more points than set one with 543 total points. Similar to the first data set the majority of the frequency dominance is in the low 0-10Hz range. However, the overall spectrum is significantly larger than the first data set, going up to a maximum frequency of approximately 275Hz. It still retains the evenly spaced peaks as the magnitude trails off; however, these are further apart, closer to 15Hz between each peak. Again, the Y axis seems to have slightly higher dominant frequencies compare to the X and Z components.

L2-Norm of 1s Future Predicted Position: 0.28

Next Steps/Ideas:

It is clear that there are similar components between the supplied data's frequency spectrums such as the location of the dominant frequencies and the peaks as the magnitude diminishes. A possible improvement could be to extrapolate the positions of these smaller peaks and ensure a frequency is included to represent these peaks in the inverse transform whilst still squashing lower frequencies. This could be done with a dynamic threshold that looks at a small portion of the spectrum and adjusts based on the local maximums to ensure these peaks remain in the data.

Additionally, there are a handful of points where the position of the point is lost by the 3D camera. These jumps to 0 may be impacting the output of the prediction algorithm. Potential smoothing of the input data could be implemented to mitigate the effect these jumps to 0 have on the output.

Investigation into additional prediction methods could be conducted to identify general models for ocean movement from which to base the prediction. Such a model could be combined with the stream of input data to generalize the prediction to better suit fluctuating conditions.

There is also potential to incorporate the motion of the ROV into the algorithm. As the ROV moves it creates relative motion of the tracked point since the camera moves with the ROV. Using the motion commands/motion of the ROV to counteract this relative motion could allow for a more accurate model of the tracked point to be developed and then recombined with the ROV motion to determine a final position.